

Making comparisons

- Groups defined by **strata** labelled in categorical variables
- **Observations** in strata, same or different?
- Is there a **baseline**, or normal value?
- What are the **dependencies** in the way the data was collected?
- Are multiple samples recorded for the same individual, or recorded on different individuals?

Determining the appropriate comparison is not always easy, but important to document when decided.

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How would you answer these questions?

- Are housing prices increasing more in Sydney or Melbourne?
- Is the rental price of the unit/apartment too high?
- Is the Alfred or Epworth Hospital better for giving birth?
- It's hot and dry today, is the risk of bushfires too high to go hiking?

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ETC5521: Diving Deeply into Data Exploration

Making comparisons between groups and strata

Professor Di Cook

Department of Econometrics and Business Statistics

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1

At the heart of quantitative reasoning is a single question:
Compared to what?

-Edward Tufte

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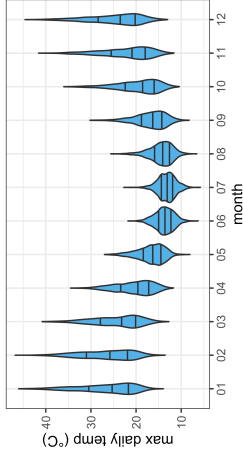
Case study: Melbourne's daily maximum temperature (1/2)



Melbourne's daily maximum temperature from 1970 to 2020.

What are the **strata** in temporal data?

- How are the temperatures different across months?
- What about the temperature within a month?



Case study: Melbourne's daily maximum temperature (2/2)

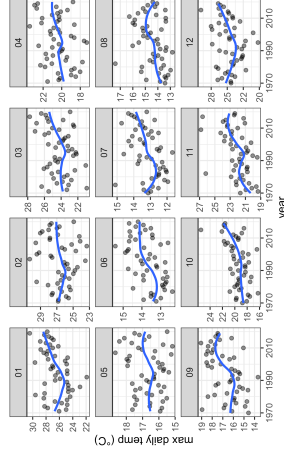
Why can we make the comparison across months?

Because it is the same location, and same years, for each month subset.

Is some variation in temperature each month due to changing climate?

How would you check this?

► Code

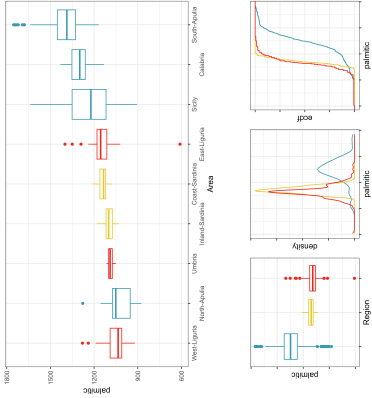


What is `scales="free_y"` for? *ED-ie*

Outline

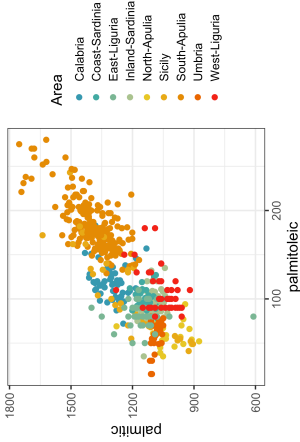
- Comparing strata
- Paired, matched or repeated measurements
- Normalising, standardising, baseline, calibration
- When there is no data
- Including uncertainty
- The problem of making thousands of comparisons
- Generating comparison samples

Comparing strata



- The olive oil data consists of the percentage composition of 8 fatty acids (palmitic, palmitoleic, stearic, oleic, linoleic, linolenic, arachidic, eicosenoic) found in the lipid fraction of 572 Italian olive oils.
- There are 9 collection areas, 4 from southern Italy (North and South Apulia, Calabria, Sicily), two from Sardinia (Inland and Coastal) and 3 from northern Italy (Umbria, East and West Liguria).

Case study: olive oils (2/4)



Colour is generally good to differentiate strata but if there are too many categories then it becomes hard to compare.

Activity: Seasonality over years

Your turn

What does change in seasonality mean?

1. Using `netlb.temp`, compute the average max daily temperature for each **year** and **month**.
2. Make a line plot of temperature by month (x-axis), with one line connecting the dots for each year. Is there any evidence that the shape of the seasonal cycle has changed over time?
3. Years differ in their overall average temperature, which can make it hard to compare the *shape* of the seasonal cycle rather than its level. Try calibrating each year by dividing its monthly temperatures by that year's average temperature, then remake the plot. Does this change what you see?
4. What does this calibration achieve, and what does it hide?

Code Learn

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► Code

Case study: olive oils (1/4)

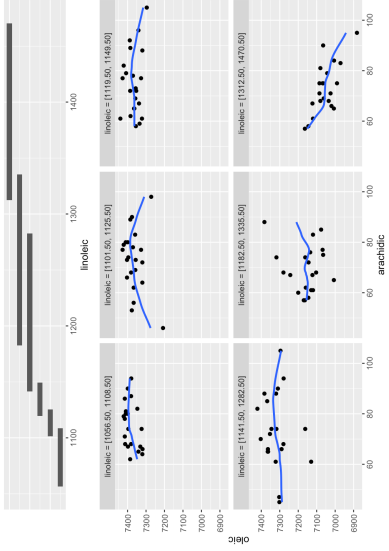


Strata from quantitative variable

► Code

The `coplot` divides the numerical variable into chunks, and facets by these. The chunks traditionally we overlap.

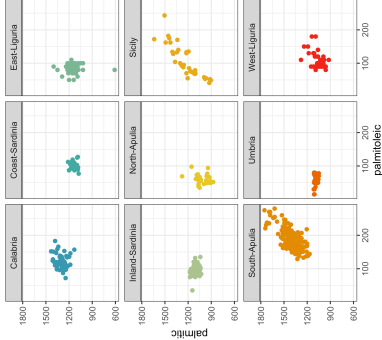
Becker, Cleveland and Shyu, (1996), Cleveland (1993)



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Case study: olive oils (3/4)

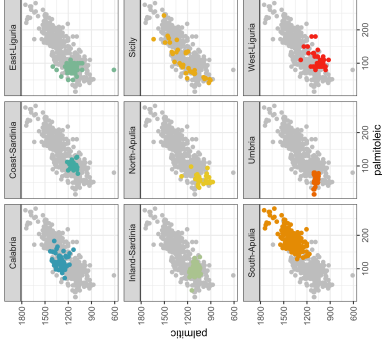


It can be hard to compare across plots, because we need to remember what the previous pattern was when focusing on the new cell.

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Strata for categorical variables



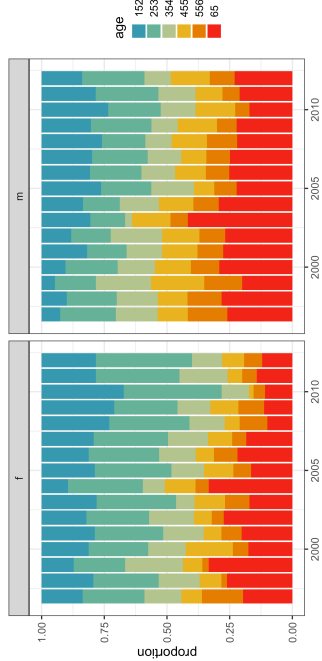
Comparison to all, by putting a shadow of all the data underneath the subset in each cell.

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Simple rearranging comparisons: tuberculosis (2/3)

► Code



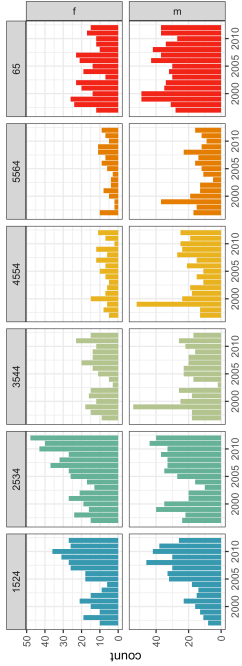
Primary comparison is **age**, relative to yearly trend.

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Simple rearranging comparisons: tuberculosis (3/3)

► Code



Primary comparison is **year** trend, separately for age and sex.

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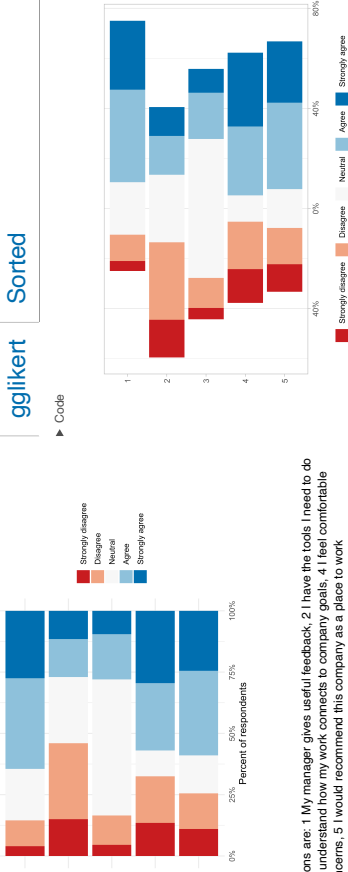
► Code

► Code

A Likert plot centres the bar on the neutral category.



► Code



The questions are: 1 My manager gives useful feedback, 2 I have the tools I need to do my job, 3 I understand how my work connects to company goals, 4 I feel comfortable raising concerns, 5 I would recommend this company as a place to work

Rank the 5 questions by “most disagreement”.

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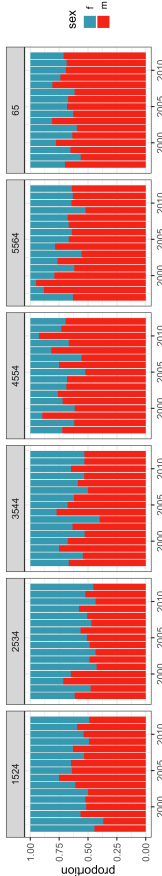
NOTE: Also a good idea to sort questions (not done here).

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Simple rearranging comparisons: tuberculosis (1/3)

► Code



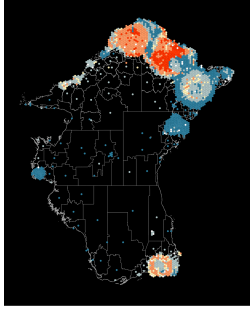
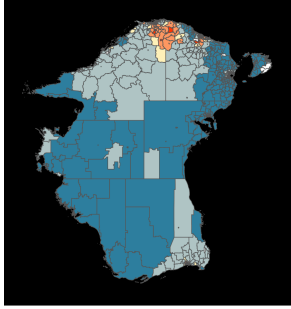
Primary comparison is **sex**, relative to yearly trend.

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Case study: choropleth vs hexagon tile ^(1/3)

The goal is to demonstrate that the *hexagon tile map* is better than the *choropleth* for communicating disease incidence across Australia.



The choropleth fills geographic regions (LGAs, SA2s, ...) with colour corresponding to the thyroid cancer relative difference from the overall mean. The hexagons, are also filled this way.

Kobakian et al

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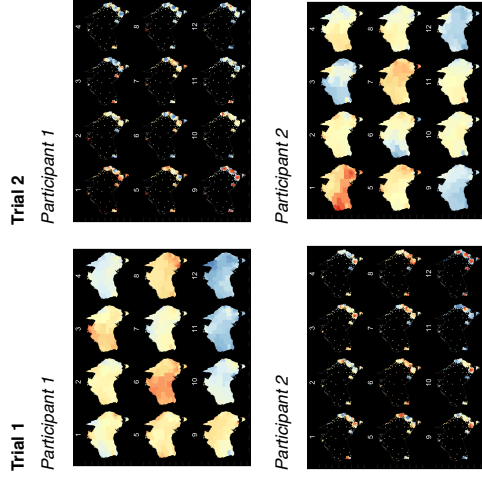
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Case study: choropleth vs hexagon tile ^(2/3)

- Each participant can **only** see the (same) **data once**.

- Need to test for different types of spatial patterns.
- Need to repeat measure each type of pattern, and each participant.

Pairing is done on the data set. Four different data sets used for each pattern.



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Paired, matched or repeated measurements

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Pairing adjusts for individual differences

If we were wanting to measure the effect of incorporating a data analytics competition on student learning which is the best design?

METHOD A

- Divide students into two groups. Make sure that each group has similar types of students, so both groups are as similar as possible.
- One group gets an extra traditional assignment, and the other group participates in a data competition.
- Each student takes an exam on the content being taught.
- The scores are compared using side-by-side boxplots and a two-sample permutation test

METHOD B

- Each student takes an exam on the content being taught. We'll call this their **BEFORE** score.
- Divide students into two groups. Make sure that each group has similar types of students, so both groups are as similar as possible.
- One group gets an extra traditional assignment, and the other group participates in a data competition.
- Each student takes an exam on the content being taught. We'll call this their **AFTER** score.
- The **difference** between the before and after scores are compared using side-by-side boxplots and a two-sample permutation test

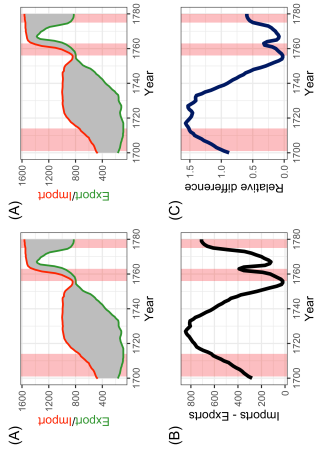
What other modifications to the design can you think of?

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Famous example: trade

data R



- The export from England to the East Indies and the import to England from the East Indies in millions of pounds (A).
- Import and export figures are easier to compare by plotting the difference like in (B).
- Relative difference may be more of an interest: (C) plots the relative difference with respect to the average of export and import values.
- The red area correspond to War of the Spanish Succession (1701-14), Seven Years' War (1756-63) and the American Revolutionary War (1775-83).

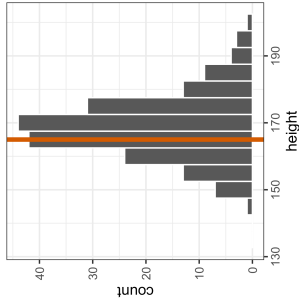
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Am I short?

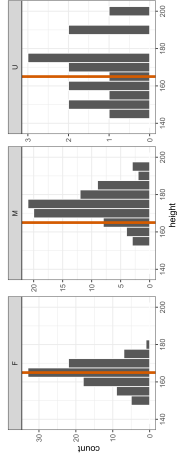
I am 165 cms tall.

► Code



But, there are strata in humans, so compared to what would be better?

► Code



Nope, I'm average height.

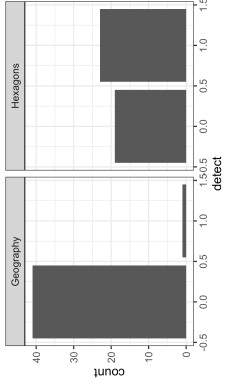
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Case study: choropleth vs hexagon tile (3/3)

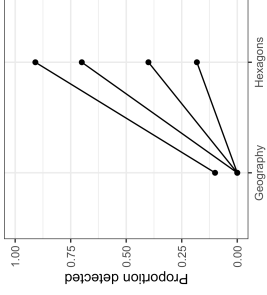
Ignore the pairing

► Code



Account for the pairing

► Code



Looks like detection rate about 50-50 for hexagon tile map, which is better than almost zero for choropleth map.

For each data set, the hexagon tile map performed better.

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Normalising, standardising, baseline, calibration

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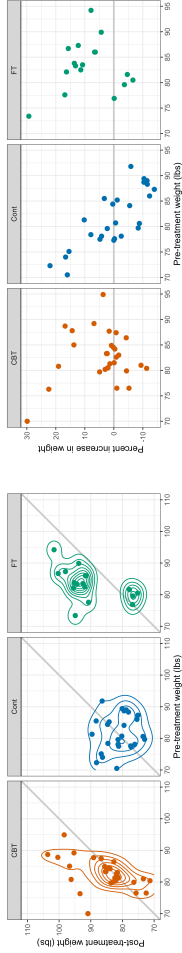
Common ways to standardise observations

Put rows on the same footing so different individuals/units can be compared

Method	Formula	Use when
Ipsatization	$x_{ij} - \bar{x}_i$	Respondents differ in baseline scale use
Row z-score	$(x_{ij} - \bar{x}_i)/s_i$	Respondents differ in spread too
Compositional %	$x_{ij} / \sum_j x_{ij}$	Compare allocation pattern, not scale
Library-size norm.	adjust for exposure/depth	Unequal measurement effort per unit
Unit norm (row)	$x/\ x\ $	Compare pattern, not magnitude (e.g. text)
Rank-within-row	rank across own items	Avoid scale-use differences entirely

Comparing with a baseline

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- Primary comparison is before treatment weight, and after treatment weight.
- Three different treatments.
- Compute the difference
- Compare difference relative to before weight
- Before weight is used as the baseline

Standardising

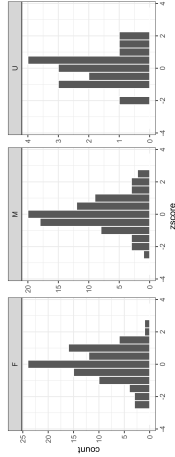
Within each strata convert values to a z-score.

- females: $\bar{x} = 164.43, s = 6.41$
- males: $\bar{x} = 174.23, s = 8.71$
- unknown: $\bar{x} = 165.74, s = 18.15$

My z-score is 0.09.

$$Z = \frac{x - \bar{x}}{s}$$

► Code



Rob's height is 170 cms. His z-score is -0.49.

I am relatively TALLER than Rob.

Range of standardising methods

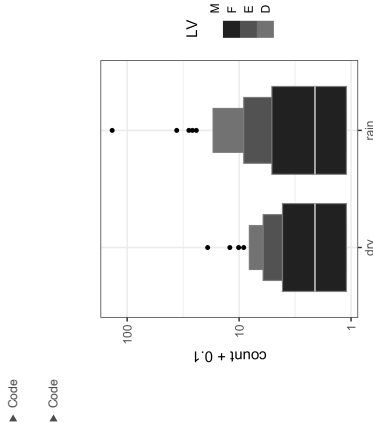
Put values on the same footing so different subgroups can be compared

Method	Formula	Use when
Z-score	$(x - \mu)/\sigma$	Roughly normal, no big outliers
Min-max	$(x - x_{min})/(x_{max} - x_{min})$	Need a bounded [0,1] range
Robust scaling	$(x - \text{median})/IQR$	Outliers / skew present
Unit vector norm	$x/\ x\ $	Care about direction, not magnitude
Rate / per-unit	$x/\text{exposure}$	Comparing groups of different size
Index / rebase	$100 \times x/x_{base}$	Comparing trends across units
Percentile rank	$\text{rank}(x)$	Compare relative standing

What happens when denominator is small with some of these metrics? ED-le

Comparison when only have positives ^(1/2)

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Are there less sightings of orchids on rainy days?

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Famous calibration error: ovarian cancer proteomics (2002–2005)

A 2002 study claimed a mass-spectrometry blood test could detect ovarian cancer with near-perfect accuracy — using mass spectroscopy to display proteins in serum as a series of peaks, with a computer algorithm finding patterns unique to patients with the disease. It generated enormous excitement (a commercial test, “OvaCheck,” was being built on it).

Keith Baggerly and Kevin Coombes at MD Anderson reanalyzed the underlying data and could not reproduce the reported sensitivity and specificity, tracing the problem not to the biology but to how the data had been collected and processed.

See [New York Times article](#).

Baggerly called this style of work *forensic bioinformatics*.

- Discriminating peaks were spread across the entire spectrum, when true cancer-related protein changes should only affect a few specific peaks — a sign the “signal” wasn’t biological.
- Some of the “important” peaks fell in regions of the spectrum where the machine couldn’t have been reliably sampling proteins at all — meaning the signal was probably just instrument noise.
- Crucially, Baggerly pointed to incomplete randomization of samples as a likely source of these problems — cancer and control samples tended to be run in different batches, on different days, so any drift in the machine’s calibration between runs became perfectly confounded with disease status. The “biomarker” was **picking up which day the sample went through the machine, not cancer**.

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When there is no data

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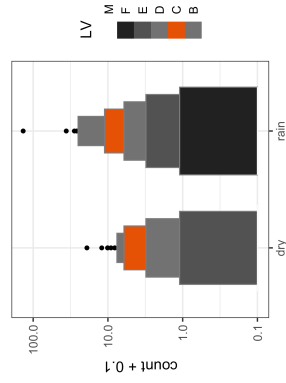
Including uncertainty in plots

Comparison when only have positives (2/2)

The usual approach: Error bars bolted on

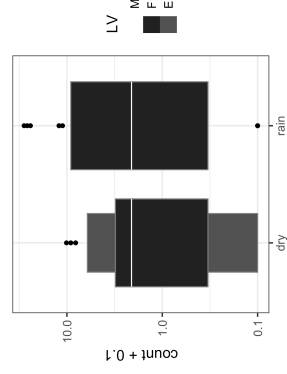
Join using the full time data as the primary set.

► Code

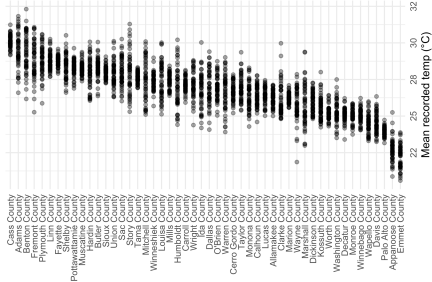


Orchids are a seasonal sighting, and there might be a difference in precipitation between seasons. *Solution:* Restrict to a month where orchids are typically visible.

► Code

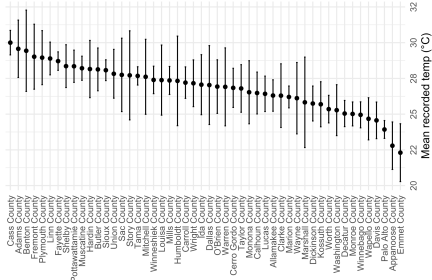


► Code



Swap the number for a `dist_normal()`, use the `_sample` geom, and `ggdibbler` draws `times` plausible realisations — uncertainty shows up as visual spread/noise, the same plot you'd already make, just fed a random variable instead of a point estimate.

► Code



Uncertainty is a *separate layer* added after the fact — easy to skip, easy to ignore.

The problem of making thousands of comparisons

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ggdibbler: Uncertainty *is* the data



Generating comparison samples

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Case study: gene expression ^(1/2)

A typical experiment tests **every gene** for a difference between conditions: often 20,000+ tests at once, each with its own p -value.

The fix: control the **False Discovery Rate**, not α used in each test

Benjamini-Hochberg procedure: rank p -values, adjust each against its rank, relative to number of tests, keep everything below the corresponding threshold.

- At $\alpha = 0.05$, **5% of truly null genes will look “significant” by chance alone**
- With 20,000 genes and (say) 18,000 truly unaffected by the treatment:

$18,000 \times 0.05 \approx 900$ false positives, even if nothing real is happening

- A single-test error rate (α) controls the wrong thing here: it bounds the false-positive rate *per test*, not across the whole gene list you'll actually report

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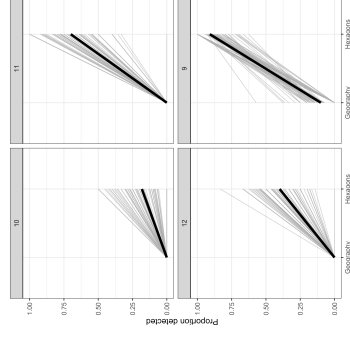
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Bootstrap confidence intervals



- Confidence intervals show what might happen to estimates with different samples,
- with the **same dependence structure**.
- Sample the current sample, but don't change anything else.
- Reason for sampling with replacement, is to keep sample size the same - we know that variance decreases with smaller sample size.

For choropleth vs hexagon tiles, **sample participants with replacement**.



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Case study: gene expression ^(1/2)

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Key points

- Compared to what is especially important for exploring **observational data** — the strata, dependencies, and baseline all need to be identified before comparing.
- Adjust for individual variation, by
 - pairing (or repeated measures on the same unit)
 - comparing relative to a baseline
 - standardising onto a common scale (z-score, rate, index, ...)
- Calibration matters: comparisons across batches, instruments, or samples can produce spurious “signals” if not properly randomised or normalised — **it’s the process being compared, not the effect of interest**.
- When data only records presence (not absence), join against the complete set of possibilities so **missing combinations count as zero, not nothing**.
- Uncertainty should be part of the estimate, not an afterthought bolted on with error bars.
- Making **many comparisons at once** inflates the chance of false discoveries — correct for multiplicity.
- Bootstrap samples and permutation lineups let you **simulate what the comparison would look like** under resampling or under no real effect, to judge whether an observed difference is real.

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Resources

- Wilke (2019) [Fundamentals of Data Visualization](#) Chapters 9, 10, 11
- Unwin (2015) [Graphical Data Analysis with R](#) Chapter 10
- UNL Statistics faculty (2025) [The Metrics: The Good, the Bad, and the Ugly](#) — rebuttal seminar on the metrics used to justify the elimination of the UNL Department of Statistics

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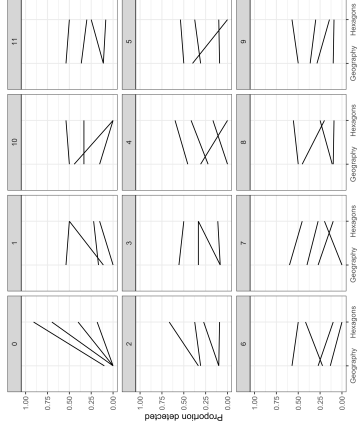
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Lineups

- Lineups show what might happen to estimates with null samples, where (by construction) there is no relationship.
- Thus you need to **break dependence structure**.

For choropleth vs hexagon tiles, randomise the type of plot each participant received. This breaks any dependence between type and detection rate.

► Code



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Activity: Comparisons gone wrong, for real

On December 5, 2025, the University of Nebraska Board of Regents voted 7-1 to eliminate the Department of Statistics at UNL entirely, along with three other departments (Earth and Atmospheric Sciences, Educational Administration, and Textiles, Merchandising and Fashion Design), as part of a \$67.4 million cut affecting 512 positions — Statistics alone accounted for 317.5 million and 12 eliminated positions. All degree programs (BS, MS, PhD) were closed, and tenured and tenure-track faculty were let go, with their last day of employment set for May 14, 2027.

The decision leaned heavily on a set of cross-department performance metrics — exactly the kind of *comparison* problems from today’s lecture.

Your turn

Look at the 7 slides under “Comparisons” in the [UNL Statistics faculty’s rebuttal deck](#).

Can you see what comparison calculation was wrong?

[?](#) [Learn](#)

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